

MPLS architecture

A typical MPLS architecture is shown in Fig. 1. Components of an MPLS IP network are customer edge (CE), label edge router (LER) and label switching router (LSR).

CE works at IP level. LER is the entry point of MPLS domain and is known as the provider edge. LSR works as a transit switch between LERs. Label switched path (LSP) is the data path between two routers through which packets travel. Lines shown between CE and LER carry IP packets bi-directionally.

Customer edge (CE). It structures the customer message into IP packets and sends it to the entry node of MPLS domain. While receiving IP packets from the egress node of MPLS domain, CE sends packets to the network layer of its own, after removing the IP address.

Label edge router (LER). This works as the gateway of MPLS domain and sits at the edge of the MPLS domain. Ingress LER receives IP packet from CE and assigns the appropriate label. After wrapping the label, it sends the labelled packet towards the next hop through LSP, which is assigned for the specific forward equivalence class (FEC). Assigning the label is termed as label binding. LER also acts as the egress router. Egress LER receives labelled IP packets from the previous transit router, pops up the label (removes the label) and routes the IP packets towards the destined CE.

Label switching router (LSR). This functions as a transit switch in MPLS cloud. It receives labelled IP packets through the appropriate LSP. It analyses the label bound over the packet, consults the forwarding information table [label information base (LIB)] and routes the packet through the appropriate LSP.

LIB is a software database created in both LER and LSR. It contains mapping information of the incoming label and LSP with outgoing label. It is created during installation of the router and, subsequently, updated

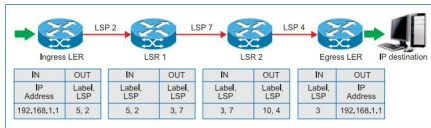


Fig. 2: Illustration of traffic flow in MPLS domain

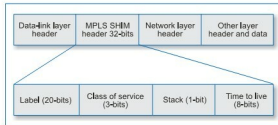


Fig. 3: Generic MPLS label format

automatically when the new LSR and LER are added by using label distribution protocol.

When LSR is routing the packets from the incoming LSP to the outgoing LSP, it strips out the incoming label and assigns a new label to the same packet to ensure security from intruders. This process is known as label swapping or label changing.

Label switched path (LSP). Within an MPLS domain, a path is set up prior to data transmission for a given packet to travel based on forward-equivalence class (FEC). There are two types of LSPs; one is static and the other is signalled. Static LSPs are configured manually on each LSR. No signalling protocol is used in static LSP. To establish a static LSP, the operator configures ingress LER, transit LSRs and egress LER, manually specifying labels to be applied at each hop.

Contrary to this, signalled LSPs are configured only at ingress LER. When a packet is assigned to a signalled LSP, it follows a pre-established path from LSP's ingress LER to its egress LER.

How MPLS works

Suppose an IPv4 packet arrives at the ingress router (LER) from CE with a destination address of 192.168.1.1

(Fig. 2). In MPLS domain, ingress router has a route for 192.168.1.0/20 with a next-hop of LSP. A header with a label of, say, 5, selected from its LIB, is appended to the packet and forwarded downstream. LER binds the selected label (5) according to the FEC over the IP packet and sends it through the pre-programmed LSP (2) towards LSR1.

On receipt of the labelled IP packet, LSR1 analyses the label only and ignores the IP address. It consults with its LIB for further routing. As a result, it removes the incoming label (5), winds the newly-assigned label (3) over the IP packet and sends it towards LSR2 over the assigned LSP (7).

LSR2 consults with its LIB and transmits the IP packet after swapping the incoming label (3) with the outgoing label (10) towards egress LER over the pre-assigned LSP (4). Egress LER stripes the label (10), goes through the destined IP address (192.168.1.1) and hands it over to the correct CE.

Forward-equivalence class

FEC is a representation of a group of packets that share the same requirements for their transport. All packets in such a group are provided with the same treatment en route to the destination. FEC assignment is based on:

1. Class of service requirement
2. Quality of service requirement
3. Prefixes of IP addresses

FEC based on class of service requirement. IP packets from different users are categorised on the basis of class of services these are entitled to,